

Project Name: Data Analysis for Supply Chain

Task Assignment & Roles - Defined responsibilities for team members:

- The project team (Data Innovators) and their assigned roles:
- Omar Shaaban Nour: Data Cleaning & EDA
- Karim Moutaz Saad: Customer Segmentation
- Hady Yaser: Product Performance Analysis
- Abdullah Ali El Shabrawy: Regional Analysis
- All team members: Sales Trends Analysis, Reporting & Visualization

Risk Assessment & Mitigation Plan

Risks & Mitigation:

- Data Quality Issues: Missing or inconsistent data.
Mitigation: Perform thorough data cleaning and preprocessing.
- Scope Creep: Adding too many analysis questions.
Mitigation: Stick to the predefined scope and prioritize key questions.
- Tool Limitations: Difficulty using Tableau or Python libraries.
Mitigation: Provide training or seek online resources.
- Time Constraints: Delays in completing tasks.
Mitigation: Allocate buffer time and monitor progress regularly.

Contingency Plan:

If a team member falls behind, redistribute tasks or extend deadlines if possible.

Key Performance Indicators (KPIs):

- Data Quality: Ensures the dataset is clean and reliable for analysis.
- Customer Segmentation Accuracy: Helps in targeted marketing and personalized customer engagement.
- Product Performance Metrics: Identifies opportunities to optimize inventory and boost sales.
- Regional Sales Growth: Highlights areas for business expansion and improvement.
- Sales Trend Accuracy: Validates the reliability of sales forecasts and insights.
- Report Quality: Ensures the final deliverables meet stakeholder expectations.
- Project Timeliness: Keeps the project on track and within deadlines.
- Visualization Effectiveness: Makes insights easy to understand and actionable.
- Actionable Recommendations: Drives real-world impact and business growth.
- Stakeholder Satisfaction: Measures the overall success of the project from the stakeholders' perspective.

Feedback & Evaluation

The lecturer's assessment of the project will focus on the following key areas:

1. Quality of Analysis: Depth and accuracy of insights derived from the data.
2. Clarity of Documentation: How well the project is documented, including the report, code, and visualizations.
3. Implementation: Effectiveness of the data cleaning, analysis, and visualization processes.
4. Testing: Validation of findings through statistical tests and peer reviews.
5. Presentation: How well the findings and recommendations are communicated to stakeholders.

Suggested Improvements

1. Interactive Dashboards:

- Create interactive dashboards using tools like Tableau or Power BI for better stakeholder engagement.
- Include filters and drill-down options for more detailed exploration.

2. Real-Time Analysis:

- Implement real-time data processing to provide up-to-date insights.

3. Stakeholder Collaboration:

- Conduct regular check-ins with stakeholders to gather feedback and refine the analysis.
- Include a feedback loop to ensure the final deliverables meet stakeholder expectations.